Issue	**01**
Factory ID	**25401**
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Velocity Textiles

F-235, S.I.T.E, Near Indus University, (Site Campus) 75700, Karachi
Karachi
Sindh Pakistan

Latitude: [24.905947], longitude: [66.988821]
Date of inspection: 29-Jan-2025

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 29-01-2025 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

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1. Executive Summary

On 29-Jan-2025, Abdul Rahim and Majid Khan on behalf of the Pakistan Accord carried out an initial electrical inspection for Velocity Textiles, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representatives Mr. Rizwan Butt (Admin Manager), Mr. Syed Mudassir (Compliance officer), Mr. Hamza (Compliance officer), Mr. Uzair (Electrical engineer) and other representatives from the factory were also present. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were also present.

The factory operates one general shift. Approximately 520 workers are employed in the factory, including 80-90 female workers.

The factory mainly comprises of 02 buildings and utilities, tenanted by Velocity Textiles.

Building 1 a five-storey (Ground+Mezzanine+3) used for offices and washing (ground floor), stitching (Mezzanine floor), stitching, checking and packing (1st and 2nd floor) and storage of goods, prayer area and solar panels (3rd floor). The building was constructed in 2000.

Building 2 a two-storey (Ground+1) used for cutting, storage, warehouse and chemical store (ground), stitching (1st floor) and solar panels (roof).

Utilities, ground only four different structures which include boiler shed, generator rooms, LT rooms and reinforced concrete ETP structure and a compressor is installed on the roof.

A summary of main concerns identified on-site during the assessment (non-exhaustive) that should be actioned are as follows:

- **Non-compliant and non-functional fire alarm system** was found during the assessment which poses significant risks. The fire alarm system has major issues, including non-compliant panels, cables, and hooters, poor visual and sound coverage, faulty loop connections causing false alarms, and broken call points. Urgent upgrades are needed for proper functionality.
- **Hot Spots** in electrical panels and distribution boards were found during the thermographic testing could be in result of several issues that may include loose connections, overloaded circuits, aging components, imbalanced loads, faulty part or poor ventilation of panels that may lead to equipment failure and fire hazards.
- **Non-compliant, insufficient emergency and directional signage** was found during the assessment lights hinder safe evacuation during power outages, increasing accident risks. Prompt repairs, added lighting, and compliance with safety standards are essential for occupant safety.

• **Lack of earthing in distribution boards and equipment** was found which increases the risk of electrical shocks, equipment damage, and fire hazards. Immediate grounding and proper earthing are essential for safety and compliance with electrical standards.

We see no reason to suspend operations in the facility due to these concerns.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view)- Red dashed line indicates site boundary, highlighted (blue) area indicated building inspected.



Figure 1.2. Site layout. highlighted area (blue) indicates buildings inspected.

2.1 Building Use

The factory consist of two main buildings and utilities only.

Building 01:
L00-Ground Floor: Offices and washing.
M00-Mezzanine Floor: Stitching.
L01-First Floor: Stitching, Checking and Packing.
L02-Second Floor: Stitching, Checking and Packing.
L03-Third Floor: Storage of goods, Prayer area and Solar panels.

Building 02:
L00-Ground Floor: Cutting ,Storage ,warehouse and chemical store.
L01-First Floor: Stitching.
L99-Roof: Solar panels.

Utilities:
L00-Ground Floor: Boiler shed, generator rooms, LT rooms and ETP.
L99-Roof: compressor.

2.2 Utility Source

The facility has an electrical supply from K.E via pole mount transformer of 250 kVA. The electrical supply from the K.E at 400V is fed to the electrical room LV panels. The supply from the backup generators and the solar PV panels is also connected to the LV panels at the electrical room. The electricity from these panels is then fed to the facility distribution boards via the electrical cables.

2.3 Backup source

The factory has 3 no. of diesel operated electrical generators for power backup. Further the emergency lights and the fire alarm system is equipped with built in batteries .

3. Documentation

S.NO	Description	Availabe (Y/N)	Created Date	Status / Remarks
1	As-Built Layout Drawing	Yes	28-12-2022	Does not comply for approval
2	Single Line Diagrams	Yes	28-12-2022	Does not comply for approval
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of Velocity Textiles, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Velocity Textiles. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Accord for approval.

Upon completion of the work, the necessary evidence required to verify that the tasks have

been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Diagrams & Drawings

5.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that electrical room layouts drawings were available however there are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202502-007522

Timeline	Remediation Plan
Medium Term FIN-0008552	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0008553	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

5.2 Single Line Diagram (SLD)

This section refers to the entire facility.
The electrical single line diagrams (SLDs) for the factory were provided however lacking compliance for approval

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components, cable rating , connected and demand load.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202502-007520

Timeline	Remediation Plan
Medium Term FIN-0008548	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0008549	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

5.3 Electrical layout drawing

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202502-007519

Timeline	Remediation Plan
Medium Term FIN-0008546	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0008547	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

5.4 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked detail in electrical layout drawings to comply for approval - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation was available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202502-007521

Timeline	Remediation Plan
Medium Term FIN-0008550	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0008551	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

6. DBs & Elec Protection Systems

6.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes specifically at the LV room panels where 95 sq.mm cable is connected to 500 A MCCB

Circuit breakers with higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202502-007512

Timeline	Remediation Plan
Medium Term FIN-0008529	Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 95 sq.mm cable is connected to 500 A MCCB at LV room.

6.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified hot spots within panels at the third floor finishing area, temperature of 82.6°C was noted. . Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202503-009963

Timeline	Remediation Plan
Immediate FIN-0012086	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.

Medium Term FIN-0012087	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature, as illustrated in the example images below. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term FIN-0012088	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating.



Figure 2.2 - Abnormal temperature at third floor finishing area distribution board.



Figure 2.3 - Abnormal temperature at third floor finishing area distribution board.

6.3 Earthing system

This relates to the entire facility.
During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

1. The earth pits were inaccessible and unsuitably located.
2. The earthing cables were not appropriately sized according to the relevant earthing standards.
3. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables including the following.

a. LV panels at electrical room.
b. LV panels at new electrical room
c. Generator room panels
d. Compressor panels
e. Building 01 third floor back process area distribution boards.
f. Building 01 washing area distribution board
g. Building 02 first floor distribution boards.
h. Boilers distribution board.

4. Equipotential bonding for the cable trays and metallic panel covers was missing throughout the facility.
5. Earthing of the electrical equipment were missing.

a. Generator room motors
b. ETP room motors
c. Compressors
d. Washing machines.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202502-007513

Timeline	Remediation Plan
Medium Term FIN-0008531	Design and evaluate the earthing system installation throughout the facility. i. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54. ii. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.

Longer Term FIN-0008532	iii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1. iv. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.



Figure 2.4 - Earth bond missing at generator room distribution baords

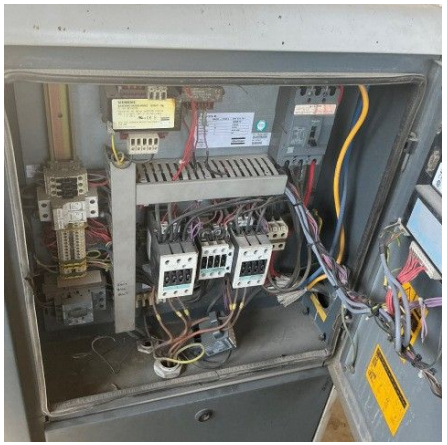


Figure 2.5 - Earth bond missing at ground floor at compressor distribution baord



Figure 2.6 - Equipotential bonding missing,tray at passage beside ETP area

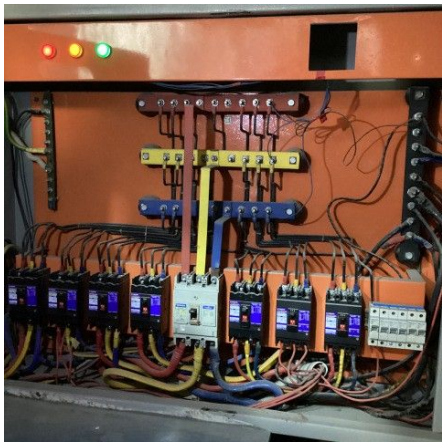


Figure 2.7 - Earth bond missing at ground floor Main Distribution Board.



Figure 2.8 - Earth bond missing at ground floor at LV room panels

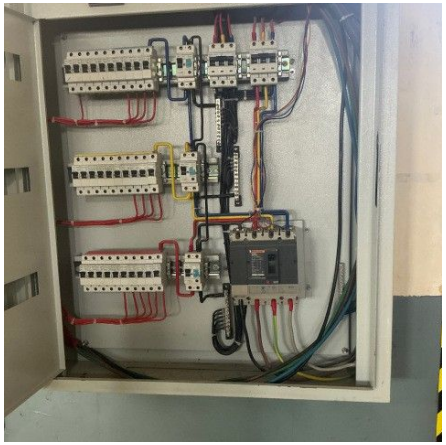


Figure 2.9 - Earth missing in building 1 , third floor back process area distribution baord.

6.4 Panel/cabling/busbar

This relates to the entire facility.
The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.
i. In a number of areas it was observed that electrical components were not suitable for the environmental conditions in which they were installed. This includes the following panels/distribution boards
a. Electrical distribution board outside ETP
b. Distribution board outside workshop for ETP
ii. Several electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements specifically the following.
a. Distribution board for compressor.
iii. There were visible signs of damage to panels, cables, and busbars. This issue was specifically observed in LV room and outside compressor room.
iv. Unmanaged cables and inappropriately installed components inside distribution board were observed at numerous locations including the following.
a. Compressor electrical connection panels
b. Electrical distribution board for compressors.
c. Boiler room distribution boards.
d. Steams generator connection panels

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202502-007518

Timeline	Remediation Plan

Medium Term FIN-0008544	i. Identify and replace electrical panels, that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022). ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials. iii. Inspect and repair or replace all damaged panels, cables, and busbars to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022). iv. Ensure proper cable management, secure terminations, correct installation of components, and adequate cable dressing per BS 7671:2018+A2:2022, Regulations 521.10.1, 526.1, 528.1, 536.4.203, and 529.1.1 to prevent mechanical strain, overheating, and non-compliance
Longer Term FIN-0008545	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.10 - Non outdoor rated distribution board at ETP area.



Figure 2.11 - Broken boards enclosure outside compressor room



Figure 2.12 - Unmanaged cables in compressor distribution board



Figure 2.14 - Incorrectly installed components and unmanaged wiring in boiler compressor distribution baord.



Figure 2.13 - Broken boards enclosure at LV room

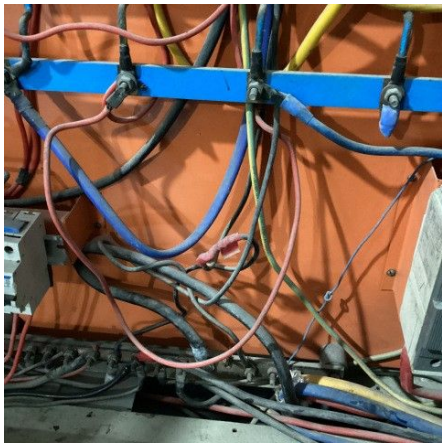


Figure 2.15 - Umanaged cables inside distribution boards at compressor

6.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system these issues were prominent at the following locations.

i. Boiler distribution board
ii. Building 02 , first floor stitching area distribution board.
iii. Cutting office panels

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202502-007515

Timeline	Remediation Plan
Medium Term FIN-0008534	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.16 - Loop connection in cutting office distribution board



Figure 2.18 - Loop connection in cutting office distribution board

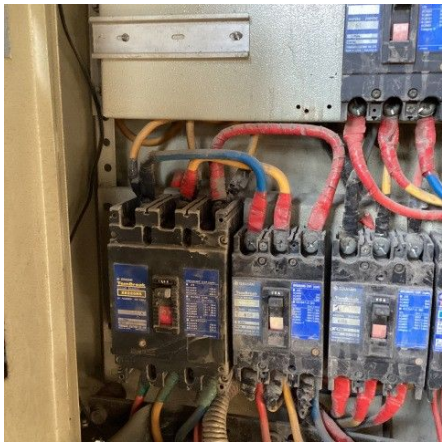


Figure 2.17 - Loop connection in boiler distribution board.



Figure 2.19 - Loop connection at building 02 , first floor in stitching area.

6.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker specifically at the following locations:
i. LV panels at the electrical room
ii. Compressor panel
iii. ETP workshop distribution board
iv. Boiler distribution board
v. Sample room distribution board
vi. Account department distribution board at ground floor
Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202502-007516

Timeline	Remediation Plan
Medium Term FIN-0008535	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.20 - Multiple looping at LV panel room ,incoming circuit breaker for utility.



Figure 2.21 - Multiple looping in boiler distribution board circuit breaker



Figure 2.22 - Multiple looping in accounts department distribution baord



Figure 2.23 - Multiple looping at circuit breaker inside LV room panels

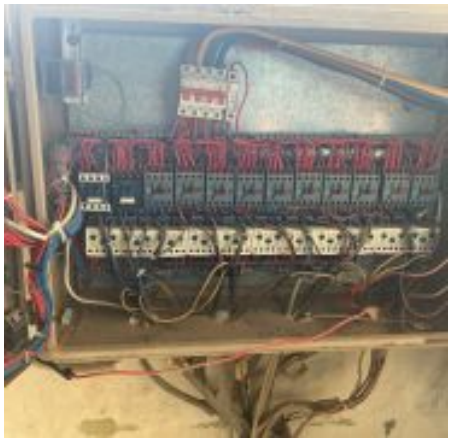


Figure 2.24 - Multiple looping in compressor panel

6.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

The inspection revealed unsealed openings in the panels and distribution boards specifically at LV room panels and washing area distribution baords.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202502-007517

Timeline	Remediation Plan
Medium Term FIN-0008536	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term FIN-0008537	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.25 - Opening at top of washing area distribution panel



Figure 2.26 - Opening in distribution board at LV room



Figure 2.27 - Opening in distribution board at LV room

6.8 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trucking systems specifically at the stitching areas through out the faciltiy.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202503-009866

Timeline	Remediation Plan
Medium Term FIN-0011851	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term FIN-0011852	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.28 - LV room distribution board contaminated by dust and debris



Figure 2.29 - Lint and dust in trays in building 02 at first floor

6.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.
The following observations were identified with the current lightning protection system:
A few lightning arrestors were installed at the factory specifically for solar panels, however no lightning protection system was installed for facility, and there was no risk assessment provided to confirm whether it was not required.
Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting F-202502-007514

Timeline	Remediation Plan
Medium Term FIN-0008533	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term FIN-0020636	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).



Figure 2.30 - Lightning arrestor installed at site

7. Fire Alarm & Detection System (FADS)

7.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

A conventional non-listed fire alarm system was installed in the factory, with the fire panel indicating multiple zone faults during the inspection. Detectors were inadequately placed, manual call points were missing in required locations, and visual/audible alarms were insufficient. Additionally, the fire alarm backup did not meet standard requirements.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm.F-202502-007585

Timeline	Remediation Plan
Immediate FIN-0008652	Ensure the functionality of the existing fire alarm system until the installation of the compliant fire alarm system.
Medium Term FIN-0008653	i. Submit design documentation for upgrading of current or new replacement FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.

Longer Term FIN-0008654	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for the import of any components required for the FADS design submitted earlier. iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation. iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review. v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation. vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and the Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.

8. Emergency Lighting

8.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations including the following:

i. Generator room
ii. New LV panel room
iii. Boiler area
iv. Ground floor warehouse
v. Fabric storage
vi. Building 02 first floor
vii. Building 01 first floor

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202502-007532

Timeline	Remediation Plan
Medium Term FIN-0008650	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.

Longer Term FIN-0008651	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.31 - Non functional exit light fixture at building 01



Figure 2.32 - Non Functional emergency lights at ground floor warehouse



Figure 2.33 - Non Functional emergency lights at ground floor building 02



Figure 2.34 - Non functional exit light fixture at new LV room

8.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting coverage. This issue was specifically observed on the ground floor of warehouse and on the first floor of building 02.
ii. Multiple emergency luminaires indicate the need for maintenance through out the facility.
iii. Installed emergency lights are too dim to provide adequate illumination thought out the facility.
iv. The backup power supply to the emergency luminaires is insufficient.
v. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during nighttime evacuations.
vi. Directional exit signage was not provided at all required locations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202502-007531

Timeline	Remediation Plan

Medium Term FIN-0008648	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation. Submit plans to Pakistan Accord for review and approval. ii. Perform maintenance on all emergency luminaires indicating issues, including cleaning, bulb replacement, and verification of electrical connections. iii. Replace dim emergency lights with higher intensity luminaires that meet the minimum illumination levels specified in the Building Code of Pakistan Fire Safety Provisions 2016. Ensure that all emergency lights provide adequate brightness to facilitate safe evacuation during emergencies iv. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016 v. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s) vi. Install exit signages at all required location to comply the standard requirement of Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term FIN-0008649	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.35 - Insufficient emergency lights in ground floor warehouse



Figure 2.36 - Insufficient emergency lights at first floor building 02

8.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202502-007530

Timeline	Remediation Plan
Medium Term FIN-0008646	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately when general lighting power fails, as required by the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term FIN-0008647	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

8.4 General Lighting

The section refers to the entire facility.

During the inspection the general lighting throughout the facility was assessed.

The following observations were noted:

i. Several locations were identified as lacking sufficient general lighting specifically at the boiler area.
ii. General lighting was controlled directly from the breakers within the Distribution Boards rather than through standard light switches through out the facility.

Insufficient general lighting in several locations can increase the risk of accidents and impede safe evacuation during emergencies, while controlling general lighting via breakers instead of light switches poses operational inefficiencies and safety hazards. This setup can delay emergency responses and heighten the risk of electrical faults or fires due to frequent breaker use. Addressing these issues is crucial for enhancing everyday safety and ensuring effective emergency preparedness in the facility.F-202502-007529

Timeline	Remediation Plan
Medium Term FIN-0008645	i. Conduct a thorough assessment of lighting throughout the facility to find areas with insufficient illumination. Install additional lighting fixtures in these spots to ensure adequate brightness. Ensure the new lighting design meets safety and efficiency guidelines to provide clear visibility for all tasks and emergency situations. ii. Reconfigure the electrical control system to ensure that general lighting is operated via standard wall-mounted light switches instead of circuit breakers. This involves installing appropriate switches and rewiring the control circuits as per the National Electrical Code (NEC) and other relevant electrical safety standards. Ensure that all switches are easily accessible and clearly labelled to enhance operational efficiency and safety.



Figure 2.37 - Lack of general lights in boiler area

9. Cabling & Wiring

9.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

The following observations revealed that the proper supports were not provided for cables, they were laid directly on the ground and hung along the walls and found hanging out of the containments. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. These observation were prominent at the following locations

a. LV panel room
b. ETP area
c. Cable trays at passage next to ETP area.
d. Generator room
e. Server room F-202502-007523

Timeline	Remediation Plan
Medium Term FIN-0008554	Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations)
Longer Term FIN-0008555	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.38 - Unsupported cables at LV room.

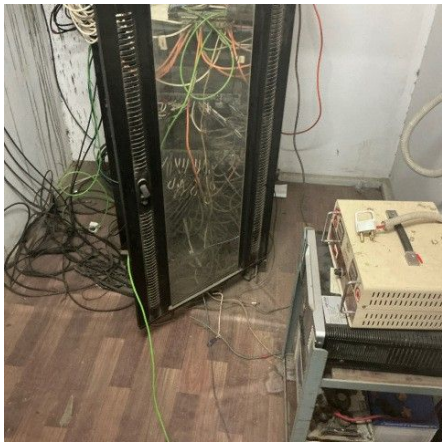


Figure 2.39 - Unmanaged cables in building 01 at first floor IT room.



Figure 2.40 - Cables falling out of containment near ETP

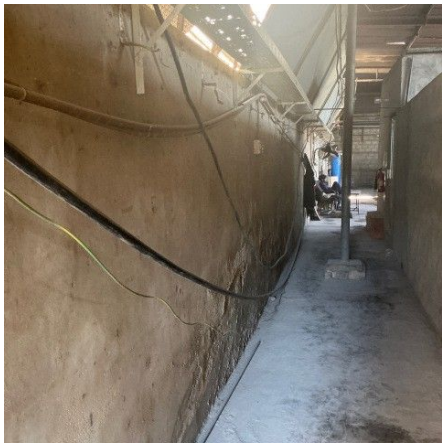


Figure 2.41 - Unsupported cables outside adjacent to passage at ETP



Figure 2.42 - Unsupported cables at LV room.

9.2 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris. F-202502-007524

Timeline	Remediation Plan
Medium Term FIN-0008556	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term FIN-0008557	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.43 - Dirt and debris in electrical room trench



Figure 2.44 - Dirt and debris in electrical room trenches

9.3 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in breaking strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed specifically at the following location:
a. LV panel room .
ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks.
a. Boiler distribution board
b. Washing distribution board on ground floor
c. Ground floor MDB
d. Generator room
iii. Machine power wires were inserted directly into power receptacles without a power plug. This practice is highly dangerous as it can lead to loose connections, electrical shocks, and short circuits. Direct insertion bypasses proper safety mechanisms, significantly increasing the risk of fire and electrical hazards, thereby endangering life safety. This issue was observed specifically at the following location:
a. Sample room on the ground floor

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety.F-202502-007525

Timeline	Remediation Plan

Medium Term FIN-0008564	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022) ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring. iii. Replace the directly inserted power wires with proper power plugs to ensure safe and secure connections.
Longer Term FIN-0008565	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections. ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques. iii. Develop and enforce a facility-wide electrical safety protocol that mandates the use of power plugs for all equipment. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards.



Figure 2.45 - Cables terminated on breakers without lugs

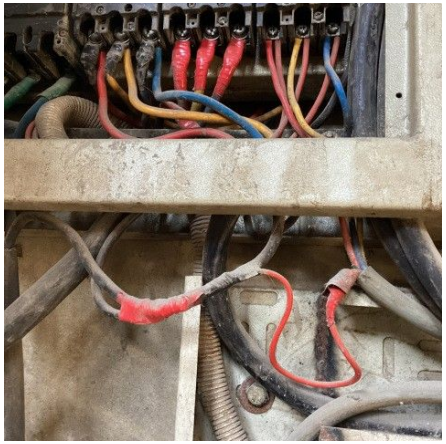


Figure 2.46 - PVC tape joints in cables at boiler distribution board

in generator room



Figure 2.47 - PVC tape joints in generator room cables



Figure 2.49 - Inappropriate cable termination at lift panel in cutting department

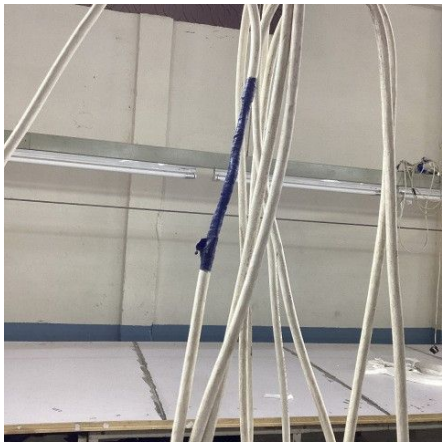


Figure 2.48 - Tape joint in machine cables at ground floor cutting department

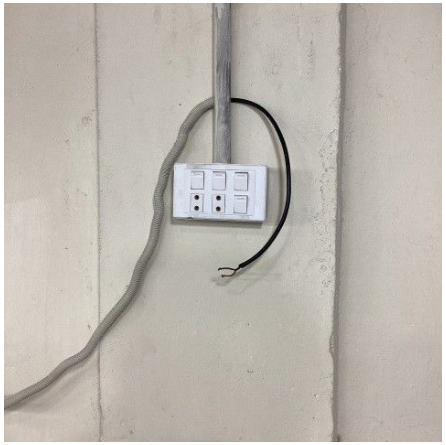


Figure 2.50 - Cables inserted into the socket without the use of plugs at ground floor sample room



Figure 2.51 - PVC tape joints in generator room cables

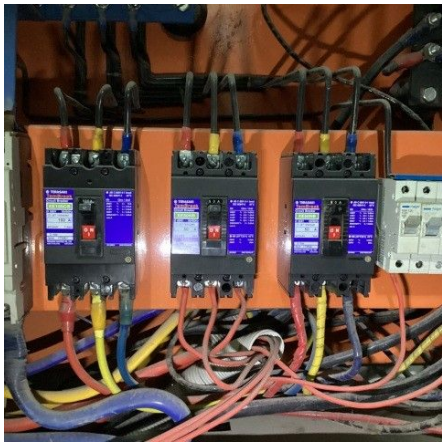


Figure 2.52 - Cables terminated on breakers without lugs in main distribution board at ground floor



Figure 2.53 - Inappropriate cable termination in ground floor distribution board

9.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable color coding practices were identified in some distribution boards specifically at third floor stitching distribution board.Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized color coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202502-007526

Timeline	Remediation Plan
Medium Term FIN-0008569	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term FIN-0008570	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.54 - Color of the cables not as per the standard in finishing area distribution board

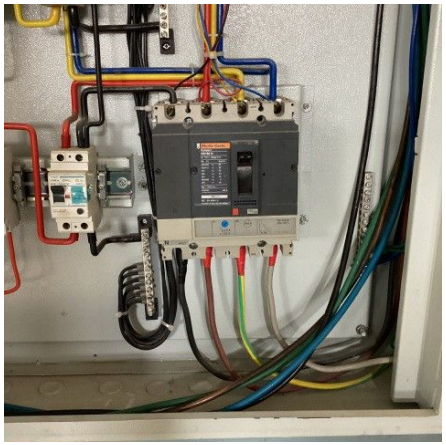


Figure 2.55 - Color of the cables not as per the standard in finishing area distribution board

9.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles installed in inappropriate locations, Receptacles placed in unsuitable locations on floor with open wire, such as non-IP rated sockets installed outdoors or exposed to the elements, are at higher risk of moisture ingress and corrosion. This can lead to electrical short circuits or fires, posing a severe safety hazard.
ii. Power receptacles and electrical switches mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety.
iii. LED lights were observed suspended by their electrical cables without proper mechanical support, particularly on the fourth floor of Building 01, posing a potential safety and installation hazard.F-202502-007528

Timeline	Remediation Plan
Immediate FIN-0008643	i. Relocate power receptacles from inappropriate locations to more suitable, protected areas to prevent exposure to environmental elements. For external receptacles utilize suitably rated IP fixtures
Medium Term FIN-0008644	ii. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk. iii. Provide proper mechanical support for the LED lights to relieve cable strain and ensure safe, compliant installation.
Longer Term FIN-0024232	i. Develop installation guidelines and training for personnel to ensure future power receptacles are installed in compliant and safe locations. ii. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance. iii. Implement standardized installation practices to ensure all lighting fixtures are mechanically supported and not suspended by electrical cables.



Figure 2.56 - Lights hanging on wire in building 01 at fourth floor

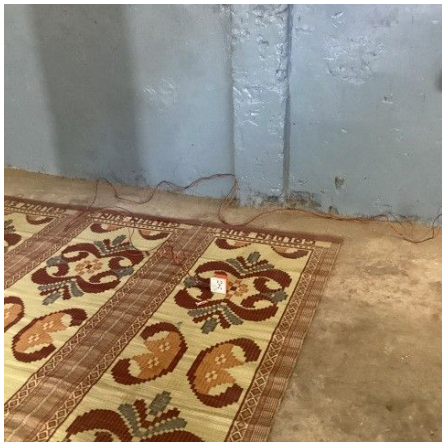


Figure 2.57 - Receptacle unsafely laid at floor

10. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.